

Network-Wide Traffic State Estimation and Rolling Horizon-Based Signal Control Optimization in a Connected Vehicle Environment

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Abstract—This paper presents an innovative method to adaptively optimize traffic signal plans based on the estimation of traffic situation achieved from the information of various penetration rates of Connected Vehicles (CVs). The network-wide signal control problem is formulated as a linear optimization problem. Moreover, we develop a Kalman filter (KF) and Neural Network (NN) algorithms to predict and update the traffic situation under mixed non-connected and connected vehicles environment. To capture the dynamic of the traffic flow, we employ the cell transmission model synched with the Vissim traffic simulator. The methodology is tested using a challenging network of six intersections. We test our model for various Penetration Rates (PR) of the CV to provide a comparative analysis. The performance of the method is also compared with a conventional actuated-coordinated traffic signal plan. The results show that with a bare minimum PR (say more than 30%), our proposed methodology outperforms the actuated traffic signal plan. (note that the minimum PR is subject to further ongoing research in the literature, to the extent that lower PRs might be plausible). Though a 100% PR is highly desirable, our method can fetch the maximum benefit just by 60% PR.

Index Terms—Connected vehicles, adaptive traffic signal control, penetration rate, Kalman filter, neural network, connected transport system.

I. INTRODUCTION

TO COMBAT traffic congestion, traffic signal setting has been a subject of extensive research, for which adaptive traffic signal control using data from fixed sensors such as loop detectors and video cameras, has been widely used [1]. However, this approach suffers from several limitations. First, fixed sensors are only able to sense the presence of vehicles in specific spots, unable to provide some insights into the overall situation of traffic across the whole network. Second, the installation and maintenance of these sensors are highly costly. Alternatively, with today's technology, traffic data can be collected from floating sensors such as mobile phones [2], GPS devices, or connected vehicles. In a connected vehicle environment, vehicles equipped with onboard units (OBUs) can communicate and exchange their information such as position, speed, acceleration/deceleration with other connected

vehicles, and similar technology installed on the roads to collect and disseminate data known as roadside units (RSUs). Moreover, the RSUs can provide drivers with information on traffic signals such as remaining green/red time or a commensurate speed to pass signal(s) within the green wave [3]. These data can provide a spatial view of the traffic situation based on which; a traffic controller can adjust signal parameters at intersections or freeway ramp meterings [4] in a smart and informed way. Moreover, this data can be used for many safety and mobility [5] applications such as cooperative hazard warning [6].

Unlike fixed sensors that are only installed in specific parts of the network [7], connected vehicles can exchange their information with RSUs installed across the network on a real-time basis. Moreover, collecting data from connected vehicles (such as speed, position, acceleration/deceleration) is relatively cheap and affordable [8], and faulty communication between some CVs or RSUs can be overlooked or corrected given the wealth of available data from the rest of the CVs. These precious data in a CV system provides the opportunity to adaptively regulate the traffic at signalized intersections.

A vast majority of the adaptive traffic signal control strategies in a connected vehicle system are tailored for a situation in which all vehicles are connected [9], [10]. Therefore, they cannot work appropriately under the situation with low PRs. This assumption is a rough exaggeration; since it will take at least 25-30 years to have 95% of vehicles with communication devices [11]. Therefore, it is of critical importance to develop traffic signal control strategies that can work for mixed traffic (i.e. various penetration rates (PR) of connected vehicles and non-connected vehicles) [6], [12].

One of the innovations of this paper is to integrate a traffic estimation algorithm into the traffic signal optimization problem to design the optimal traffic signal even in the situation with low market penetration rates of CVs. First, this paper develops a method to estimate the traffic state (i.e. the residual number of vehicles and flow) based on the data derived from a limited number of connected vehicles in mixed traffic based on Kalman filter (KF) and Neural Network (NN) techniques. Then the estimated traffic condition (in terms of the number of vehicles in each cell) will be used in a rolling horizon optimization problem to determine the optimal signal plan for the next time-step. The main contributions of the paper are four folds: first, a prediction algorithm is devised to estimate the traffic for low PRs and then integrated into an optimization

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problem. Secondly, considering the real-time traffic situation, the network-wide control strategy is designed. Thirdly, the optimization problem is converted to linear optimization and designed to be solvable in real-time. Fourthly, a rolling horizon approach is applied to consider some steps ahead of the traffic situation.

The rest of the paper is organized as follows: in the next section, we provide a review of the existing adaptive traffic signal control strategies designed for connected vehicle ecosystems. Section 3 is dedicated to the problem formulation. Section 4 presents the methodology including the traffic estimation and the optimization problem followed by the formulation and the solution method in Sections 5 and 6 respectively. We discuss the numerical results in Section 7. Concluding remarks are also discussed in Section 8.

II. LITERATURE REVIEW

The literature concerning the traffic signal control strategies may be divided into two main categories: (i) methods developed for an isolated intersection [13]–[16] and (ii) strategies considering the interactions between adjacent intersections [9], [14]. The main advantage of the isolated traffic signal control strategies is the computational efficiency, which comes at the expense of failing to consider the interaction of adjacent intersections, resulting in the suboptimal performance of control algorithms [17].

Network-level control strategies seek to design optimal signal plans for a network of adjacent traffic signals [18]–[23], considering their endogenous interactions aiming to provide better traffic circulation (less congestion). This paper also concentrates on the network-wide signal setting; hence the rest of the literature review also focuses on the network-wide control strategies. However, traffic signal control in the level of the network is an NP-hard problem, that is, the computational time for real-life examples becomes a prohibitive factor [24].

Given the methods used, a vast majority of the research in the past can be divided into two categories (i) centralized [20] (ii) and hierarchical [18], [25] traffic signal control strategies. In the former, all the involving parameters (i.e., cycle length, green split and offset) are identified using a single optimization problem. Whereas, in the hierarchical methods the parameters are identified via a top-to-bottom approach. For example in [18] at first, the optimal green time for each phase is identified which is then carried over to a different optimization problem to optimize the offset in the corridor.

Generally speaking, centralized methods are superior to the hierarchical counterparts, in a sense, all the parameters are found in one go. Hence the interaction between them is not lost which is the case in the hierarchical method. Moreover, given the nature of the hierarchical method, resorting to a bilevel (or multi-level) programming problem is inevitable [18]. This would add an extra layer of computational burden, that bilevel problems are in essence proven to be NP-hard [26].

It is preferable to consider optimizing traffic signal plans in a given time horizon into the future, which is the mandate

of an approach, namely, the rolling horizon [10], [20], [25], [27], [28].

To take into account, the future situation of the traffic in the signal setting (i.e. providing the signals with a series of phasing for say every 5 minutes in the future), one may need to solve the signal timing problems (centralized or hierarchical methods) for multiple time-steps in the future [29], [20]. The main idea is to set up a signal plan at a given time-step (say “*i*”) while considering the traffic conditions of the next time-steps in the future (say *i*+1, *i*+2, *i*+3 ...). As a result, it takes the dynamic nature of the problem into account and prevents a myopic determination of signal timing. Otherwise, it leads to the suboptimal performance of the algorithms. For example, it is intuitively plausible that for a globally optimal solution a particular movement receives maximum red time to avoid congestions in the next time-steps.

The rolling horizon method suffers from a detrimental issue called the “curse of dimensionality”. In a rolling horizon optimization problem, the total number of decision variables comes out of the number of time-steps multiplied by the number of effective variables in each time-step (i.e., cycle length, green split and offset), which, again, raises the concern of computational burden. To reduce the number of decision variables in the rolling horizon strategy, we restrict the number of control variables to one that only decides about the right of way for a particular movement at each intersection for the next time step. Therefore, in our methodology, there is no need to determine cycle length, green split, and offset at each time step.

It is also important to discuss the issue of penetration rate (PR) of the connected vehicles [30]. The PR can significantly affect the availability of information about the traffic situation, and in turn, it gets reflected in the performance of signal settings. The majority of the previous algorithms consider all or a significant proportion of vehicles to be connected (i.e. PR = 100%) [31]–[33]. However, based on the survey carried out by Volpe National Transportation Systems Centre [11] it will take between 25-30 years to reach a penetration rate of 95%.

Although the performance of the traffic signal control strategies highly depends on the availability and accuracy of the traffic data, only a few of the research studies consider estimating the traffic states for the situation with an imperfect penetration rate of CVs [34]–[36]. Hence it is important to develop signal control algorithms tailored to mixed traffic scenarios (i.e. connected and non-connected vehicles), especially when there exists a limited number of connected vehicles (low PR). To this end, some studies try to project the data collected from the connected vehicles to non-connected vehicles. For instance, Feng *et al.* [25] designed a method to estimate the location and the speed of non-connected vehicles based on their interactions with connected vehicles. Then these data are used in the design of the signal control algorithm. One of the limitations with this method is that it cannot estimate the location of non-CVs in the free flow condition when there is no interaction between CVs and non-CVs. The prediction algorithm cannot estimate the vehicle locations accurately (in terms of prediction error) when the percentage

of connected vehicles to the total number of vehicles is less than 25%.

He *et al.* [20] developed a simple regression model to identify the platoons using Vissim simulator under various penetration rates and traffic conditions in a mixed CV and non-CV environment. However, the root mean square error of the estimation algorithm increases quickly when the penetration rate of connected vehicles starts to decrease from 40%.

Lee *et al.* [37] applied a Kalman filter technique to estimate the cumulative travel time in the situation with imperfect market penetration rates of CVs. Based on [38] the Kalman filter outperformed different methods in traffic status estimation. Based on the paper's report, at least 30% of CVs are needed to achieve the benefits.

In this paper, we develop an innovative traffic estimation algorithm which is integrated with a signal control strategy to adaptively control the traffic based on real-time traffic condition. Therefore, the main contribution of the paper is twofold: first, using the estimation algorithms to better estimate the number of vehicles in each cell even in the situation with low market penetration rates of connected vehicles. Second, integrating the estimation results into a rolling horizon network-wide traffic signal optimization algorithm to optimally regulate the traffic in different situations. The proposed optimization method considers some steps ahead of the traffic situation to design the traffic signal at each time step. Moreover, it also adaptively circulates the traffic in various traffic conditions by considering a flexible cycle length and sequence of phases.

The estimation algorithm has the following characteristics:

- It fuses a Kalman filter and Neural network algorithm to respectively predict the flow and the residual number of vehicles at each time step
- It does not track the trajectory of individual vehicles and therefore the vehicles' privacy is truly addressed
- The algorithm can estimate the cell densities in different traffic conditions and for the whole network and not only in the vicinity of the intersections
- Based on the results of this paper, the critical penetration rate for a beneficial performance of the estimation algorithm is 20% which is comparable with other existing studies

Given the shortcomings of previous studies, this paper proposes an adaptive traffic signal control strategy which aims to address the followings:

- In the situation with imperfect market penetration rates of CVs, an estimation module is formulated as the feedback for the optimization problem to increase the efficiency of the optimization
- Using the estimation algorithm, traffic signal control can fetch the maximum benefit just by 60% PR
- To prevent suboptimal performance, the network-wide control strategy is developed based on the rolling horizon optimization
- The signal control optimization is formulated as a mixed-integer linear optimization problem with a limited number of control variables

- The method is implemented on a real-time basis for a test network of intersections

The literature on signal control strategies in the presence of CVs are summarized in Table I. Different assumptions and attributes about the penetration rate, optimization approach, and metric, estimation method, baseline signal control strategy, as well as traffic model attributes, are compared for various studies. Based on the results most of the control strategies assumed the PR of CVs to be known and to be high (some consider 100% PR). A few numbers of studies focus on developing a method to estimate the non CV information from the CV data. Moreover, most of the control strategies are non-centralized approaches designed for an isolated intersection. The control algorithms with coordination are mostly developed and tested for corridors (not a network of intersections). Only a few of them are tested and developed for a network of intersections

The baseline scenarios for the majority of literature are either actuated or fixed time signal plans. The rolling horizon is also utilized in a few numbers of studies in a CV system.

The contribution of this study is twofold: first, using the estimation algorithms (NN and KF), to better estimate the traffic states even in the situation with low market penetration rates of connected vehicles. The novelty factor is to develop a fusion method based on the NN and KF which estimates the total cell densities based on CV data. Moreover, to the best of our knowledge, only a limited number of studies are developed to estimate the cell states which is important for the accurate performance of the cell transmission-based adaptive signals. Besides, the penetration rate is estimated to choose the right class of NNs and KFs associated with each PR. Second, the estimation results are integrated into a network-wide signal optimization algorithm based on a rolling horizon strategy which considers some steps ahead of the traffic situation in the design of the signal plan. Moreover, considering variable cycle length and sequence of phases in the control strategy adds to the flexibility of the optimization problem to determine the optimal solution.

III. PROBLEM FORMULATION

This paper seeks to adaptively control the traffic signals at the network-level based on real-time data from connected vehicles. First, the estimation algorithm is applied to specify the congestion (in terms of the number of vehicles in each corridor) in various parts of the network. Then a cell transmission-based rolling horizon optimization strategy is developed to determine the next time step optimal traffic signal plan for the entire network. The objective function aims to maximize the intersection output of all intersections for some time steps ahead into the future. The decision variables will determine the optimal signal plan of all traffic signal controllers. The optimization problem is formulated as follows:

Objective function = Maximizing the intersection output of all intersections for some time steps ahead into the future

Subject to the following constraints:

- Flow conservation model
- Conflict movements constraint
- Restriction on maximum allowable waiting time

TABLE I
REVIEW OF SIGNAL CONTROL STRATEGIES IN THE CV SYSTEM

Ref	Penetration rate (known/estimated)		Penetration rate			Centralized/non-centralized	Rolling horizon	Optimization metric	Non CV position estimation	Baseline	Model attributes
	Known PR	Estimated PR (sensors)	High (>50%)	Low (20-50%)	Very low (1-20%)						
[9]	✓		✓			Non-centralized	×	Waiting time and queue length (not an optimization problem)	×	Random control Best-first control	Intersection
[10]	✓		✓			Non-centralized	✓	Maximize the intersection output and minimize the queue length	×	Coordinated fixed time Actuated coordinated	Network
[14]		✓	✓	✓	✓	Non-centralized	×	Algorithm to minimize the total delay or number of stops	✓ Arrival time prediction	First in first out	Intersection
[15]	✓		✓			Non-centralized	×	Minimize delay	×	Actuated/Webster/Pretimed	Intersection
[16]	✓		✓			Non-centralized	×	Density (not an optimization problem)	×	An online algorithm	Arterial
[39]	✓		✓	✓		Non-centralized	×	Waiting time (not an optimization problem)	×	Actuated plan	Intersection
[25]		✓	✓	✓		Non-centralized	✓	Minimize total vehicle delay and queue length	✓ Location & speed estimation	Fully actuated control	Intersection
[40]	✓		✓	✓	✓	Non-centralized	✓	Delay, stop, and deceleration	×	Coordinated actuated	Intersection
[31]	✓		✓			Non-centralized	×	Control delay/ Queue length	×	Pretimed/adaptive	Intersection
[41]	✓		✓	✓	✓	Non-centralized	×	Accumulated delay	×	COLOMBOs approach	Intersection
[33]	✓		✓			Non-centralized	×	-	×	Pretimed	Intersection
[18]	✓		✓	✓		Hierarchical	×	Minimize individual & platoon delay		Actuated-coordinated control	Corridor
[19]		✓	✓	✓	✓	Centralized	✓	Total delay/ total queue length minimization	✓ Location & speed estimation	Actuation (Act) +priority	Intersection
[20]	✓		✓	✓		Centralized	✓	Total weighted delay	✓ Platoon identification	Actuated/coordinated/priority	Corridor
[22]		✓	✓	✓		Non-centralized	✓	Minimize the total queue length	✓ Queue (ql-estimation)	TRANSIT-7F	Network
[23]	✓		✓			Non-centralized	×	Queue length	×	VANET-based/ Webster	Network
[32]	✓		✓			Non-centralized	×	Density	×	Pre-timed	Intersection
[37]		✓	✓	✓	✓	Non-centralized	×	Cumulative travel time	✓ Cumulative travel time (KF)	Actuated signal control	Intersection

IV. METHODOLOGY

Let us first introduce all the notations and abbreviations used in this paper (see Table II).

The framework of our methodology is based on the estimation of traffic states (i.e., the number of vehicles in each cell) and the rolling horizon optimization strategy. The cell transmission model (CTM) [42], which is a widely utilized traffic model [43], is used to determine the traffic state of the next time-steps given signal settings of the current time-step.

However, in our optimization strategy, to better respond to the real traffic situation, only the signal setting of the first time-step is realized, and the rests are discarded.

Based on the methodology developed in this paper, the number of vehicles in each cell is updated at the beginning of solving the optimization problem. To do so, we utilize the information of CVs captured in the previous time-step to update the number of vehicles in the respective cell based on an estimation algorithm consists of a neural network (NN) and Kalman filter (KF). The NN method is used to calculate the residual vehicles. Moreover, a KF model is also developed to calculate the inflow/outflow of vehicles to/from each cell being derived from the Vissim traffic simulator. As a result, a combination of the NN and KF methods renders a more reliable estimation of the initial state of the cells which is then used as an input in an optimization problem ($n_c(t)$).

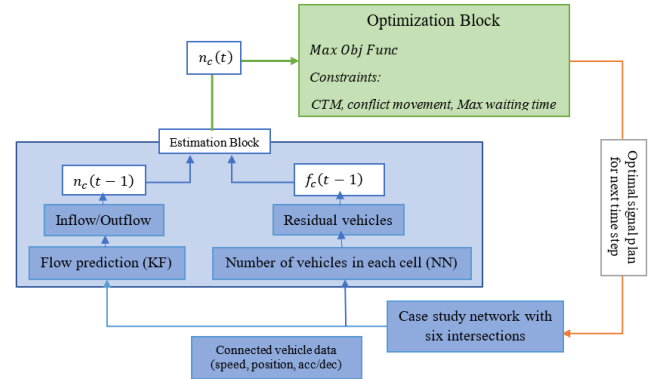


Fig. 1. Traffic signal control strategy.

The details of the NN and KF techniques are explained in the next subsections. Fig. 1 outlines the methodology of this paper which consists of two main blocks 1) Estimation Block 2) Optimization Block. As can be seen, at the outset, the CV data (such as position, speed, acceleration/deceleration) will be used in the estimation block (NN and KF models) to determine the residual number of vehicles as well as the inflow/outflow of vehicles to/from each cell. This data will then be used as an initial value ($n_c(t)$) for the optimization block, in which, the CTM specifies the cell states for the

TABLE II
NOTATIONS AND ABBREVIATIONS USED IN THIS PAPER

Sets	
$i \in I$	Set of all intersections
$c \in C$	Set of all cells
C_m	Set of all middle-cells
C_i	Set of all intersection-cells
C_{ib}	Set of all intersection- before -cells
C_{ia}	Set of all intersection- after-cells
C_{so}	Set of all source-cells
C_{si}	Set of all sink-cells
T	Total number of time-steps in the rolling horizon
M	Set of all conflict movements at each intersection
Symbols	
$n(i, j, t)$	Number of all vehicles in cell j of intersection i at time-step t
Decision variable	
$\theta(t)$	Binary variable indicates the signal status at each time-step t
Variables	
y_p	Yellow time of phase p
r_p	Red clearance time of phase p
T_m	Maximum number of times that an approach can receive red
$n_c(t)$	Number of vehicles at cell c at time-step t
$N_c(t)$	Maximum existing capacity at cell c at time-step t
$f_{ic}(t)$	Inflow to cell c at intersection i at time-step t
$Q_c(t)$	Maximum possible inflow from cell c at time-step t
Abbreviations	
ARIMA	Autoregressive moving average
CV	Connected vehicle
CTM	Cell transmission model
KF	Kalman filter
MAPE	Mean absolute percentage error
MOE	Measure of effectiveness
NN	Neural network
OBU	Onboard unit
PR	Penetration rate (of connected vehicles)
RMSE	Root mean square error
RSE	R-Squared
RSU	Roadside unit

intermediate time-steps inside the prediction horizon based on the values obtained from the estimation block. Therefore, $n_c(t)$ only used as the initial value for the CTM to be able to update the states in the next time steps inside the prediction horizon. The optimization block seeks to maximize the traffic output of all intersections for some time steps ahead into the future. As a result, an optimal signal setting can be regarded as a sequence of signal plans maximizing the intersections output during a pre-specific time horizon into the future. Moreover, to update the traffic signal plan based on recent data, conservatively, only the first signal timing setting will be realized.

This process will be repeated for a sufficient number of iterations while observing some measures of effectiveness (MOEs) such as travel time, delay, stop delay, and emission. In the end, we compare the proposed algorithm against an optimized actuated-coordinated signal plan.

Subsection A dedicated to the estimation algorithms. In subsection B, we elaborate on the theory of rolling horizon optimization.

A. Traffic States Estimation Using KF and NN

In this section, we explain the performance of the estimation algorithms (NN and KF) used in our methodology.

The short-term prediction of the traffic states (such as flow and queue) is an inseparable part of adaptive traffic signal control strategies. Moreover, the performance of the traffic signal algorithms is highly correlated with the accuracy of the prediction results. The real-time predicted data which describes the traffic situation used as input to determine the parameters of adaptive traffic signal controllers [44]. In this section, we propose a prediction method based on data obtained from connected vehicles.

The traffic flow and the residual number of vehicles are calculated based on KF and NN respectively. Considering the nonlinear nature of the queue estimation problem, the NN model is chosen to be used for the residual number of vehicles estimation. The reason behind this choice is that the NN model can represent the nonlinear relationship between input and output variables and it has been widely used as a method-of-choice [45]–[49]. Moreover, it is intuitively conceivable that connected vehicles (as probe vehicles) represent the traffic circulation. As the increase/decrease in the flow of connected vehicles has a direct relationship with the increase/decrease in the total flow. Despite the nonlinear nature of the queue estimation because of the shockwave phenomenon, the relationship between the flow of CVs and the total flow has a linear nature for which we use a KF model. Based on this model, total flow is represented based on the flow of CVs and white noise. The main premise of the KF model is the linear correlation between the components which is easily generalizable for different traffic situations. Moreover, the linear Kalman filter demands less computational cost in comparison with its nonlinear counterpart. The KF is a popular method of flow prediction in the literature [50]–[52].

Following, these algorithms are explained in detail.

1) Traffic Flow Prediction Based on Kalman Filter Theory:

In this part, we propose a method to estimate the inflow of vehicles to each cell from its immediately downstream cell based on the data gathered from connected vehicles. A Kalman filter is utilized to predict the total inflow based on the flow pattern of connected vehicles at each time interval. We develop an algorithm to predict the flow for various penetration rates of connected vehicles, different traffic signal control settings at intersections, and an altered pattern of inflow vehicles to the network. The KF uses a recursive procedure to estimate the flow of vehicles to each cell at each time-step based on the flow of connected vehicles and the total flow of vehicles in the previous time step. To apply the KF method for traffic flow prediction, the Kalman equations should be calibrated using different traffic conditions.

Here the Kalman filter equations are used to represent the dynamics of the traffic inflow to each cell. The first equation (1.a) is the state equation which represents the relationship

between the flow of vehicles in each time-step with the flow in the previous time-step. Here f_c is the state variable and demonstrates the inflow to each cell at time-step t . $\phi(t)$ is a 1×1 state transition matrix that relates each state to its previous state. $w(t)$ represents the noise with zero mean and covariance $Q(t)$ and normal distribution [53].

$$f_c(t+1) = \phi(t) f_c(t) + w(t) \quad (1.a)$$

$$f_{CV}(t) = H f_c(t) + v(t) \quad (1.b)$$

Equation (1.b) illustrates the measurement equation used to calculate the flow of vehicles based on the flow pattern of connected vehicles. In this equation, $f_{CV}(t)$ represents the number of connected vehicles that enter a cell at each time-step. H has a scalar value and maps the flow of vehicles to the flow of connected vehicles and $v(t)$ is the measurement noise with the same characteristics but uncorrelated with $w(t)$. To predict the flow at each time-step, the Kalman filter algorithm utilizes the previous state and the new measurement.

To build the Kalman filter equations, we need to identify ϕ_k , H , w_k and v_k in the state and measurement equations. In this study, the data gathered from the Vissim traffic simulator for different traffic situations will be used to set the parameters. f_k indicates the inflow of vehicles to each cell and can be affected by various variables, such as the traffic signal plan of the downstream intersection, the layout of the intersection, and the pattern of inflow vehicles that enter the network. We run the Vissim traffic simulator for the test network for different vehicle inputs and various fixed and random traffic signal plans for the traffic signal controllers. This can produce dissimilar inflow patterns for each cell. During the run of the Vissim traffic simulator, the traffic flow data for all vehicles, and connected vehicles that approach each cell will be collected via the Matlab COM interface. The Kalman filter parameters are then identified separately for each penetration rate of connected vehicles.

In this study, we assume a 10 seconds time interval for inflow prediction. To identify the state transition ϕ_k and noise w_k , which are considered to be scalar in this study, a linear regression model is used to identify the relationship between flow at each time-step and the flow at the previous time-step. The same procedure is done to identify the scalar H and v_k in the measurement equation based on the relationship between the total flow and the flow of connected vehicles. After calibrating the Kalman filter equations, they can be utilized in a recursive procedure to estimate the flow at each time-step.

In this paper, the flow information, as well as the residual number of vehicles in each cell, are the input variables for the optimization problem.

The residual vehicles at each cell are predicted based on the neural network technique which is explained in the next subsection.

2) Residual Number of Vehicle Estimation Based on Artificial Neural Network: To estimate the number of vehicles in each cell at each time-step, additional to the inflow of vehicles, we need to know the residual (initial) number of vehicles at each cell at the previous time-step. Our proposed

method only uses the information of connected vehicles to estimate the residual vehicles in each cell and unlike the similar queue length estimation methods, it does not need additional information about the arrival pattern of vehicles or the parameters of traffic signal controllers such as cycle length, green split or offset. Similar to the method utilized for the Kalman filter strategy, at first the NN structure needs to be trained based on data gathered from the Vissim traffic simulator. To collect data for training and testing the NN structure, the Vissim simulation data is obtained for a variety of traffic situations. Then 70% of the data are randomly chosen to train the neural network and the remaining 30% are used to evaluate the final structure of the model.

To develop an NN model that can estimate the number of residual vehicles in different traffic situations, we generate dissimilar traffic conditions such as the ones used to train the KF structure. Generating diverse traffic conditions results in a various number of residual vehicles at each sampling time-step in each cell. Therefore, the NN structure will be trained based on a variety of training data and therefore can be used in various traffic conditions. A separate NN is trained for each penetration rate of connected vehicles.

The number of connected vehicles, average speed, average acceleration/deceleration, and the standard deviation of the acceleration/deceleration of connected vehicles are considered as the inputs variables for the NN structure. Note that we consider the standard deviation of the acceleration/deceleration as an input for the NN model since the average acceleration/deceleration would be zero both at the free flow and if an equal number of vehicles accelerate and decelerate at identical rates; however, these traffic states are quite different. Therefore, considering the standard deviation of the acceleration/deceleration as an input for the NN structure would enhance the accuracy of the estimation model. The output would be the total number of vehicles in each cell at each time-step ($n_c(t)$).

After preparing the training data, the model is developed by using the nntool in Matlab software. The feed-forward back propagation neural network is chosen in this study which can estimate the data with a highly non-linear relationship between input and output variables [54]. In our application, four neurons which are equal to the number of predictors (number of connected vehicles, average speed, average acceleration/deceleration, and the standard deviation of the acceleration/deceleration of connected vehicles) are considered for the first layer. The output would be the predicted total number of vehicles. Considering the trade-off between the computational effort and accuracy, we figured out 15 neurons can estimate the number of residual vehicles with good accuracy and reasonable computational time. During the training period, the weights and biases in different layers are adjusted.

The trained NN then can be utilized to estimate the residual number of vehicles in all cells.

As noted, the NN and KF models have been trained based on a range of CVs' penetration rates. Hence our proposed algorithm needs to estimate the average CVs' penetration rate in the network to call appropriate NN/KF models. The penetration rate is calculated by dividing the number of connected vehicles to the total number of vehicles. One may argue against

the plausibility of calculating the total number of vehicles. To this end, there are several ways to obtain the total count of vehicles. One is to use the historical data which are available for various times of the day and days of a week in most cities. Such historical data is easily available from the traffic data set on official traffic websites. These historical data represent the congestion trends (in terms of the total number of vehicles) at different times of the day. The historical data ought to be compiled from various sensors/sources such as video cameras, Bluetooth sensors, and loop detectors already installed in the cities.

In case of the absence of such historic data, one can estimate traffic count from mobile phones' Bluetooth signal (some companies are collecting and providing such data at a very affordable cost, e.g. Intelmatics (<https://www.intelematics.com/connect/>)).

Therefore, it is safe to say that the denominator of the PR ratio (i.e. traffic count of the urban roads) is given and known. The signal receiving from the CV vehicle will be the nominator of the ratio, hence the PR of the CVs can be calculated.

The results of both KF and NN algorithms are used to estimate the number of vehicles in each cell at the beginning of the optimization problem. One may argue that why we need to use both of these algorithms to determine the cell state. If we specify the number of vehicles based on the inflow/outflow and without considering the initial number of vehicles in that cell, this can result in the accumulation of error in the final results of the total number of vehicles. This degradation can also be intensified over time. Therefore, using the NN method along with the KF strategy can provide a more accurate estimation. The final number of vehicles at the beginning of the optimization problem will be calculated using both the inflow from the Kalman filter and the residual number of vehicles from NN in the previous time step.

Besides, an accurate estimation of the total vehicles cannot be achieved if we only use the NN method. This is mainly because the CV data used as the input for the NN model cannot be a good representation of the total number of vehicles in low PRs. In the situation with a limited CVs, the number of connected vehicles is not sufficient to reflect the behavior of all vehicles, which results in an inaccurate estimation of the number of vehicles. Therefore, fusing data from two different algorithms can provide the optimization problem with more precise data about the traffic situation, which then results in the better adjustment of traffic signal parameters. The outcomes of the estimation algorithms (KF and NN) will be used to update the initial number of vehicles in each cell before solving the optimization problem. Based on equation (2), the number of vehicles in cell c at time t ($n_c(t)$), is equal to the residual number of vehicles in the previous time-step ($n_c(t-1)$), (calculating from NN method) plus the inflow to that cell between time $t-1$ and t ($f_c(t-1)$), (calculating from KF method) minus the inflow to cell $c+1$ between time $t-1$ and t ($f_{c+1}(t-1)$), (which again obtained from the KF method).

$$n_c(t) = n_c(t-1) + f_c(t-1) - f_{c+1}(t-1) \quad (2)$$

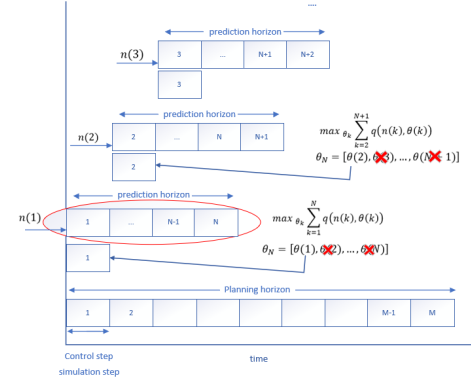


Fig. 2. The concept of rolling-horizon optimization.

It is important to note that NN and KF are two general models to estimate the residual number of vehicles and traffic flow tailored for individual cells. In the absence of real CV data, we inevitably used a micro traffic simulation software (i.e. Vissim) to generate a database. Traffic simulation packages including Vissim are skewed toward specific simulation models. Nevertheless, such limitation doesn't compromise the fidelity of our proposed methodology, in the sense that the NN and KF only procure input data. In anticipation of accessing the real CV data (as part of the ongoing AIMES project [55]) in the near future, the Vissim software will become obsolete. Since the NN and KF models do not depend on any predetermined model or parameter and also have a range of adjustable parameters, they can easily be trained for various real networks with different driving behavior.

B. Rolling Horizon Optimization Strategy

Before we write the math formulation of the optimization problem, we first explain the system dynamics. Fig. 2 illustrates the concept of rolling horizon optimization. This strategy is used to determine the optimal traffic signal setting in this paper. Consider that the aim is to determine a sequence of control variables (say $\theta_M = [\theta(1), \theta(2), \dots, \theta(M)]$) by maximizing an objective function (i.e., $q(n(k), \theta(k))$) in Fig. 2) over the whole planning horizon which consists of M time-steps. $\theta(k)$ represents the optimal control variable (signal setting in this paper) for time-step k and the objective function describes the intersection output of all intersections. Moreover, $n(k)$ shows the state of the system in time-step k . It employs the optimization problem to determine the optimal control variable $\theta(k)$. A mathematical representation of the system is also used to describe the system dynamics and determines the states of the system.

In this paper, the cell transmission model by Daganzo [42] is employed to describe the system dynamic model. One major drawback of using the dynamic model in the optimization problem is that the model used to describe the dynamic of the system may be influenced by external noise and disturbances and therefore becomes faulty.

Now assume that an updated measurement for state variable n is available at the beginning of each time-step k . Therefore, the optimization problem will be solved again starting from

the newly measured state n as the initial state. This can result in determining the control variables with more accuracy considering the real-time situation of the system. Feedback of the new measurements to the optimization problem can make the optimization more robust to external disturbances and uncertainties of the model. Therefore, the whole process can be interpreted as a closed-loop system. The rolling horizon optimization aims to determine the optimal control variables for the next time-step while considering the effect of this determination on some steps ahead of the system. The optimization problem is solved at the beginning of each time-step whenever a new measurement state ($n(k)$) becomes available. In this study, the state of the system represents the total number of vehicles at each cell. At the beginning of each simulation step, the current cell states are estimated and then fed to the optimization problem. Then the objective function is optimized for a finite time-horizon into the future which is called the prediction horizon. The prediction horizon consists of several control steps. The planning horizon represents the total number of control steps in a simulation run. Each control step is considered to be 10 seconds. At the beginning of each control step (10 seconds), the optimization problem determines the optimal traffic signal plan for the whole prediction horizon (3- control steps (30 seconds)). However, only the traffic signal decision of the first time-step (first 10 seconds) will be implemented and the rests are discarded, and it moves to the next iteration to do the same.

One may argue that why we choose the three steps ahead in our rolling horizon strategy. In the rolling horizon technique as the number of time steps ahead increases, the computational time increases significantly. This study aims to develop a method with reasonable computational time and applicable for real-time implementation in the test network. We choose the number of time steps so that the computational time be less than the duration of each control step (10 seconds) hence the next time step of the signal plan can be determined in real-time. Thus, we can implement our algorithm on a real-time basis in the simulation environment. The 3 steps ahead are chosen for this research study that results in a reasonable computational time (computational time for both estimation and optimization algorithms is about 4 seconds). Almost the same horizon plan is considered in other research studies. For instance, in [20] a rolling horizon is solved every 30 seconds for 4 cycles. Moreover, to ensure the real-time capability of the algorithm, 2 stages ahead are considered for the rolling horizon algorithm in [25]. In the next section, the cell transmission model used in this study is explained with more details.

C. Cell Transmission Model

Based on what we described for the rolling horizon optimization, we need to have a dynamic model of the system. Here we use the flow conservation model proposed by Daganzo [42]. The flow conservation model defines the dynamic of the traffic in each cell formulated as a set of recursive equations. Equations (3) and (4) are a general form of flow conservation. Equation (4) implies that the number of vehicles in each cell at time-step $t + 1$ depends on the

number of vehicles in that cell at previous time-step t , plus the inflow to the cell c between time t and $t + 1$ ($f_c(t)$), minus the inflow to cell $c + 1$ between time t and $t + 1$ ($f_{c+1}(t)$). Moreover, equation (4) specifies the inflow to each cell is restricted by 3 terms. The first term ($n_{c-1}(t)$) indicates the number of vehicles in cell $c - 1$ which is located immediately before cell c . $Q_c(t)$ signifies the maximum allowable inflow which can enter cell c during one time-step t . The last term $N_c(t) - n_c(t)$ represents the existing capacity of cell c by subtracting the number of existing vehicles from the maximum capacity. Equation (4) indicates that inflow to each cell cannot exceed any of the aforementioned three terms as written below:

$$n_c(t + 1) = n_c(t) + f_c(t) - f_{c+1}(t) \quad (3)$$

$$f_c(t) = \min \{n_{c-1}(t), Q_c(t), N_c(t) - n_c(t)\} \quad (4)$$

To make the model applicable for this study, we generalize the cell transmission model to four different types of cells.

To generalize the flow conservation theory, we consider four different types of cells, middle-cells C_m , source-cells C_{so} , sink-cells C_{si} , and intersection-cells C_i , the intersection-cells also consist of two sub-groups 1) intersection before-cell C_{ib} and 2) intersection after-cell C_{ia} . The location of these cells is represented in Fig. 3. In this section, we generalize the definition of the cell transmission model for each type of cell.

Note that equation (3) and (4) are general to ensure the flow conservation for all types of cells, these equations can be directly used for middle cells, however, there is a need for some minor amendments to make the general form applicable for source-cells, sink-cells, and intersection cells. The required changes for source and sink cells are discussed below:

- To represent the inflow to source-cells, the term $n_{c-1}(t)$ in equation (4) should be replaced by the number of vehicles entering the network from the respective source-cell at time-step t
- To represent the outflow from end-cells, the term $N_c(t) - n_c(t)$ in equation (4) which describes the existing capacity in cell c , needs to be replaced by an infinite capacity. This means the outflow from end-cell ($f_{c+1}(t)$) is restricted only by two terms, the number of vehicles in the respective end-cells and the maximum allowable outflow at time-step t .

To make the cell transmission model applicable to intersection cells, we need to consider the status of the traffic signals (green or red). As can be seen in Fig. 3, the status of the traffic signal in the intersection-before-cells can restrict the inflow/outflow of vehicles in both intersection after-cells/before-cells. For instance, the outflow from an intersection before-cell at each time-step depends on the status of the traffic signal in that cell. If the traffic signal receives green in time-step t , the outflow would be equal to $f_{c+1}(t)$; otherwise, it would be zero. The same is true for inflow to the intersection-after-cells.

It is worth noting that we disregard the length of the intersection premise, that is, we assume a vehicle that leaves an intersection before-cell enters instantly to an intersection after-cell. The generalization of the cell transmission model to the intersection middle cell, before cell and after cell is represented in equation (5).

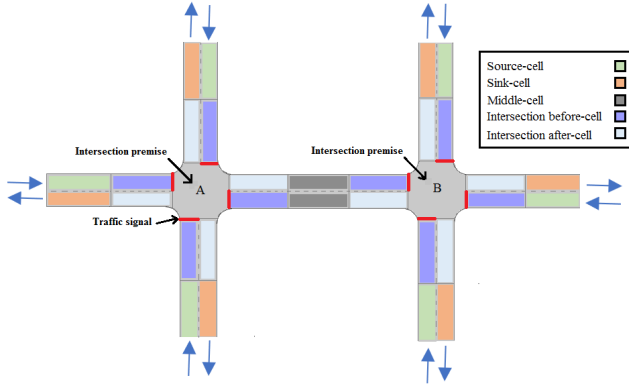


Fig. 3. The layout of the two consecutive intersections.

Note that $\theta_c(t)$ in equations (5) is a binary decision variable that is identified by solving the optimization problem at each step. This variable indicates the signal status at each approach of each intersection for the next time-step. Hence, $\theta_c(t)$ has the following definition:

$$\theta_c(t) = \begin{cases} 1 & \text{If approach receive s green for time step } t \\ 0 & \text{If approach receives red for times } t \end{cases}$$

$$n_c(t+1) = n_c(t) + f_c(t) - f_{c+1}(t) \quad (5-1a)$$

$$f_c(t) = \min \{n_{c-1}(t), Q_c(t), N_c(t) - n_c(t)\} \quad (5-1b)$$

$$c \in C - C_i$$

$$n_c(t+1) = n_c(t) + f_c(t) - f_{c+1}(t)\theta_c(t) \quad (5-2a)$$

$$f_c(t) = \min \{n_{c-1}(t), Q_c(t), N_c(t) - n_c(t)\} \quad (5-2b)$$

$$c \in C_{ib}$$

Considering the turning movements, only the intersection after-cells would be affected. To generalize the CTM for intersection after cells, the turning rates should be known for all intersections in the network. Historical data for various times of the day can be used to determine the turning movements at different intersections and various times of the day.

$$n_c(t+1) = n_c(t) + f'_{c_{ib-k}}(t) - f_{c+1}(t) \quad (5-3a)$$

$$f'_{c_{ib-k}}(t) = \sum_{\substack{ib=1 \\ ib \neq K}}^4 \alpha_{ib-k, ia-K} f_{c_{ib-k}}(t) \theta_{c_{ib-k}}(t)$$

$$f_c(t) = \min \{n_{c_{ib-k}}(t), Q_c(t), N_c(t) - n_c(t)\} \quad (5-3b)$$

$$c \in C_{ia-K}$$

In equation (5-3a), $\alpha_{ib-k, ia-K}$ represents the turning movement from intersection-before-cell ib at approach k to intersection-after-cell ia at approach K . We number the intersection approaches based on Fig. 4. We aim to obtain the flow for the intersection-after-cell at the K^{th} approach of the intersection.

The proposed methodology is summarized in Fig. 5. At first, the simulation is run for sufficient time to populate the network, a process called the warm-up period. Then initial states of the cells (in terms of the number of vehicles) are estimated based on the connected vehicle data (i.e., position, speed, acceleration/deceleration) and estimation algorithms,

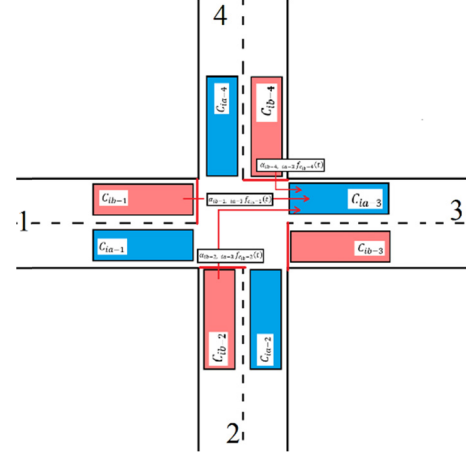


Fig. 4. Rate of turning movements at a sample intersection.

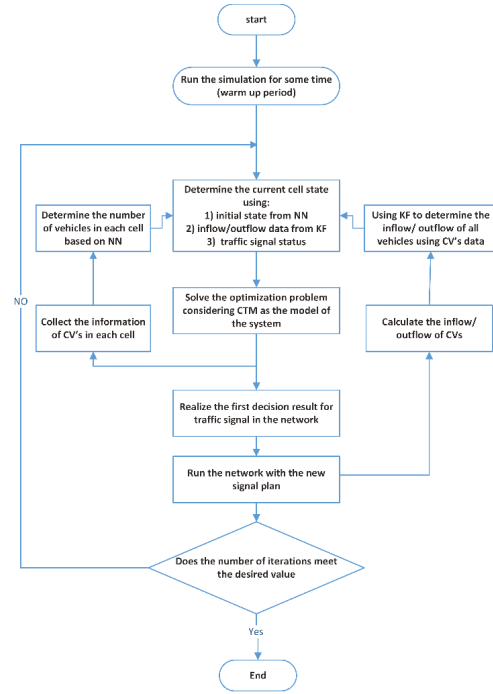


Fig. 5. A pseudo-algorithm for traffic signal optimization.

followed by solving an optimization problem to determine the signal setting for the next time-step.

Before realizing the optimal signal plan in the network, the initial number of vehicles in each cell is calculated based on the NN model. It is then completed by realizing the optimal signal settings for all intersections for the next time-step. Afterward, the KF is used while running the simulation, to determine the inflow to each cell based on the CV data. The data out of the KF, the initial state of each cell obtained from NN, and the optimal traffic signal plan information are deployed to determine the current state of the traffic. This current state is considered as the input for the optimization problem, and this cycle is repeated for a sufficient number of times.

To prevent the problem of accumulation of errors in the prediction problem, the NN technique is utilized to update the initial state of all cells at each time-step. The fusion of the NN and KF methods can provide the status of each cell accurately even when the penetration rate of connected vehicles is low.

In the next section, we elaborate on the optimization problem.

V. FORMULATION

To formulate the traffic signal plan as a rolling horizon optimization problem, we need to define an appropriate objective function and constraints which is made as a zero-one (binary) dynamic programming problem. A rolling horizon approach is adopted to maximize the intersection outputs every 10s for three steps ahead into the future. However, only the decision found for the first time-step of the optimization problem is enacted. The control variables determined from solving the optimization problem decide only on the allocation of green time to the most congested movement at each intersection at each time step. The conventional rolling horizon approaches are laden with too many signal parameters (i.e., cycle length, green split and offset) for each intersection and they do not necessarily update the signal parameters at each simulation time step (rather the control update time step can be defined to trade-off optimality and computational burden). In comparison with conventional approaches, the proposed rolling horizon strategy in this study updates the signal plans more frequently based on the new measurements of the traffic states which results in more adaptive signal optimization based on the real traffic situation, this is owed much to the fact that the size of the optimization problem is tractable for the proposed network. To make the method applicable for large networks, the network can be decomposed to some subnetworks.

The result of the proposed optimization problem specifies the next state of all traffic signal controllers in terms of binary outputs (if the decision variable for an approach is 1, the respective approach receives green, otherwise receives red). A formulation for maximizing throughput (traffic flow) of all intersections in the rolling horizon manner, is formulated in equation (6) followed by the optimization constraints in equations (7), (8), and (9).

In the objective function, $f_{c+1,i}(t)$, $c \in C_{ib}$ represents the outflow of each approach of intersection i from cell c . The aim is to maximize the outflow of all approaches of all intersections for T time-steps toward the future. The flow conservation model is used to describe the dynamic of the system. The control variables are a sequence of manipulated inputs which represent the signal status for all intersection movements.

$$\begin{aligned} & \text{Max}_{\theta_i^m} \sum_{i=1}^I \sum_{c=1}^{C_{ib}} \sum_{t=1}^T f_{c+1,i}(t) \\ & c \in C_{ib} \end{aligned} \quad (6)$$

Constraints:

Flow conservation model for different cells:

$$n_c(t+1) = n_c(t) + f_c(t) - f_{c+1}(t) \quad (7-1a)$$

$$\begin{aligned} f_c(t) &= \min \{n_{c-1}(t), Q_c(t), N_c(t) - n_c(t)\} \\ c &\in C - C_i \end{aligned} \quad (7-1b)$$

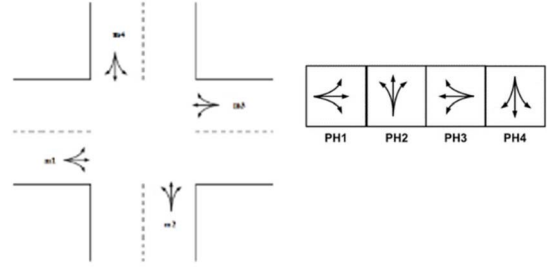


Fig. 6. Conflicting movements at each intersection.

$$n_c(t+1) = n_c(t) + f_c(t) - f_{c+1}(t) \theta_c(t) \quad (7-2a)$$

$$\begin{aligned} f_c(t) &= \min \{n_{c-1}(t), Q_c(t), N_c(t) - n_c(t)\} \\ c &\in C_{ib} \end{aligned} \quad (7-2b)$$

$$\begin{aligned} n_c(t+1) &= n_c(t) + \sum_{\substack{ib=1 \\ ib \neq K}}^4 \alpha_{ib-k, ia-K} f_{c_{ib-k}}(t) \theta_{c_{ib-k}}(t) \\ &\quad - f_{c+1}(t) \end{aligned} \quad (7-3a)$$

$$\begin{aligned} f_c(t) &= \min \{n_{c_{ib-k}}(t), Q_c(t), N_c(t) - n_c(t)\} \\ c &\in C_{ia-K} \end{aligned} \quad (7-3b)$$

Conflict movements:

$$\sum_{m=1}^M \theta_i^m(t) = 1 \quad \forall i \in I \quad (8)$$

Equation (8) ensures that only and at least one conflict movement at each intersection receives green at each time-step. In this study, we consider four different conflicting movements as shown in Fig. 6.

In equations (8), M indicates all conflicting movements at each intersection. $\theta_i^m(t)$ is a binary variable that shows the state of each traffic signal controller related to each conflicting movement at time-step t .

Restriction on the maximum allowable waiting time:

$$\sum_{t=1}^{T_m+1} \theta_i^m(t) \geq 0 \quad (9)$$

To restrict the waiting time of vehicles at each approach of the intersection, the maximum allowable waiting time of each conflict movement can be restricted by considering a limit for the number of times that an approach can receive red continuously, this can be specified by equation (9), where T_m describes the maximum number of times that a movement can receive red continuously.

Since the optimization problem has nonlinear constraints with integer control variables it is a mixed-integer nonlinear problem (MINLP). To solve it, in the next section we convert the problem to a linear optimization problem, hence the above problem is called the Original MINLP, (O-MINLP). In the next section, we present a solution method for the optimization problem.

VI. SOLUTION METHOD

As stated in the previous section, the problem of finding the optimal traffic signal plan for the next time-steps is converted

to a binary optimization problem. The result of solving this problem entails solutions for the binary variables, indicating the next status of the traffic signal for all conflicting movements in all intersections for the next time-steps. Any increase in the number of intersections or the number of time-steps in the optimization problem can increase the number of control variables which results in much higher computational time and complexity of the problem.

To solve the optimization problem and code it in MATLAB, the nonlinear flow conservation constraint, which used to describe the dynamic of the traffic flow in equation (3) and (4), [56] needs to be replaced by three additional constraints [57] :

$$f_c(t) \leq n_{c-1}(t) \quad (10a)$$

$$f_c(t) \leq Q_c(t) \quad (10b)$$

$$f_c(t) \leq N_c(t) - n_c(t) \quad (10c)$$

This conversion can result in a phenomenon called the holding-back problem, as a solution of (10a-10c) may not necessarily result in the same as of the solution of (4). Based on (10a-10c), the flow does not utilize the maximum available capacity. To prevent this problem, we also need to meet the following condition:

$$f_c(t) = \text{Max } f_c(t) \quad \text{subject to condition}(10a - 10c) \quad (10d)$$

To consider the condition (10d) in the optimization problem we need to change the objective function to the following form [58] :

$$\text{Max}_{\theta_i^m} \sum_{i=1}^I \sum_{t=1}^T \left(\sum_{c=1}^{C_m} f_{c+1,i}(t) + \sum_{c=1}^C f_{c,i}(t) \right) \quad c \in C_{mb} \quad c \in C \quad (11)$$

Moreover, in the flow conservation constraint (equation 7-2a) and (7-3a), terms $f_{c+1}(t)\theta_c(t)$ and $f_c(t)\theta_{c-1}(t)$ are bilinear products that consist of two variables. The first term ($f(t)$) is a discrete variable and describes the outflow at time t . The second variable ($\theta(t)$) is binary and represents if an approach receives green (i.e., $\theta(t) = 1$) or red (i.e., $\theta(t) = 0$) at time step t . The multiplication of these two variables forms a nonlinear term (in terms of both continuous and binary variables) and consequently causes the problem not to be mixed-integer linear programming. Therefore, we need to explicitly define the multiplication using logic constraints [59]. If the approach receives green the outflow should be equal to $f(t)$ otherwise it should be zero. To do so, let us add the following convention:

$$\theta(t) = 0 \quad \text{implies} \quad f(t)\theta(t) = 0$$

$$\theta(t) = 1 \quad \text{implies} \quad f(t)\theta(t) = f(t)$$

To linearize the nonlinear term, we replace $f(t)\theta(t)$ with an auxiliary variable $z(t)$ and consider the following constraints in the optimization problem (M is a sufficiently big value):

$$-M.\theta(t) \leq z(t) \leq M.\theta(t) \quad (12a)$$

$$-M.(1 - \theta(t)) \leq f(t) - z(t) \leq M.(1 - \theta(t)) \quad (12b)$$

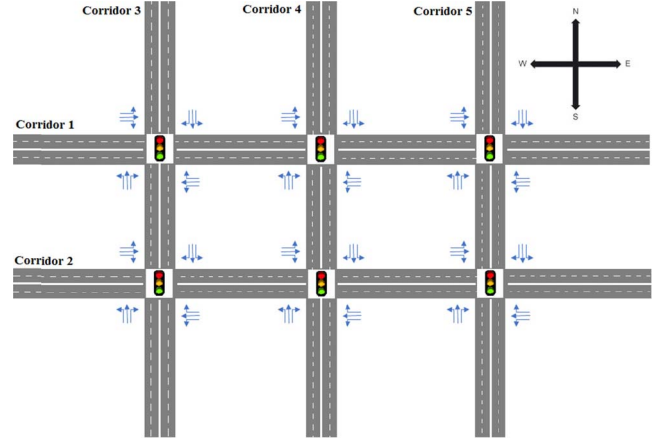


Fig. 7. The layout of the case study network with six intersections.

which infer the following:

$$\theta(t) = 0 \quad \text{implies} \quad z(t) = 0$$

$$\theta(t) = 1 \quad \text{implies} \quad z(t) = f(t)$$

These changes convert the nonlinear constraints to linear ones. Therefore, the problem alters from O-MINLP to a mixed-integer linear problem (MILP). This new formulation is called original linear MILP (OL-MILP). Table III summarizes the formulation of the O-MINLP and the OL-MILP. The OL-MILP is the final optimization problem that will be solved to find the optimal signal plan in this paper.

In the next section, the results of the proposed method are compared for various penetration rates of connected vehicles.

VII. SIMULATION RESULTS

The test networks in the previous studies mostly consist of a few numbers of intersections. Moreover, they mainly consider an arterial rather than a network. In this study, the performance of the proposed algorithm is tested on a network with six intersections built in Vissim traffic simulator. The characteristics of the study network are summarized in Table IV, and the layout of the network is shown in Fig. 7.

We analyze the result of our proposed control strategy for a test network layout in the Vissim traffic simulator with six intersections. At first, the network is simulated in Vissim. Then the real CV data are used to train the estimation methods (NN and KF). After that, the cell densities are utilized in the optimization problem to determine the optimal timing plans. Fig. 8 compares the predicted flow from KF with the actual and for two different penetration rates of connected vehicles for a random cell.

Table V summarizes the statistics related to the performance of the Kalman filter for different penetration rates of connected vehicles.

Based on the results provided in Table V, as the penetration rate of connected vehicles increases the Kalman filter method fits the data with more accuracy (R-squared) and less error (RMSE, MAPE). However, the model still has enough accuracy for a low market penetration rate of connected vehicles (say 20%). The performance of the neural network model for

TABLE III
THE FORMULATION OF THE O-MINLP OPTIMIZATION PROBLEM VS. THE OL-MILP OPTIMIZATION PROBLEM

The O-MINLP optimization formulation		The OL-MILP optimization formulation	
$ax_{\theta_i^m} \sum_{i=1}^I \sum_{c=1}^{C_{ib}} \sum_{t=1}^T f_{c+1,i}(t)$	(6)	$Max_{\theta_i^m, z_i^m} \sum_{i=1}^I \sum_{t=1}^T \left(\sum_{\substack{c=1 \\ c \in C_{mb}}}^{C_m} f_{c+1,i}(t) + \sum_{\substack{c=1 \\ c \in C}}^C f_{c,i}(t) \right)$	(11)
$(t+1) = n_c(t) + f_c(t) - f_{c+1}(t)$	(7-1a)	$n_c(t+1) = n_c(t) + f_c(t) - f_{c+1}(t)$	(12-1a)
$\zeta(t) = \min\{n_{c-1}(t), Q_c(t), N_c(t) - n_c(t)\}$ $\equiv C - C_i$	(7-1b)	$f_c(t) \leq n_{c-1}(t)$ $f_c(t) \leq Q_c(t)$ $f_c(t) \leq N_c(t) - n_c(t)$ $c \in C - C_i$	(12-1b)
$(t+1) = n_c(t) + f_c(t) - f_{c+1}(t) \theta_c(t)$	(7-2a)	$n_c(t+1) = n_c(t) + f_c(t) - z_{c+1}(t)$	(12-2a)
$\zeta(t) = \min\{n_{c-1}(t), Q_c(t), N_c(t) - n_c(t)\}$ $\equiv C_{ib}$	(7-2b)	$f_c(t) \leq n_{c-1}(t)$ $f_c(t) \leq Q_c(t)$ $f_c(t) \leq N_c(t) - n_c(t)$ $c \in C_{ib}$	(12-2b)
$(t+1)$	(7-3a)	$n_c(t+1) = n_c(t) + z_{c_{ib-k}}(t) - f_{c+1}(t)$	(12-3a)
$n_c(t) + \sum_{\substack{ib=1 \\ ib \neq K}}^4 \alpha_{ib-k, ia-K} f_{c_{ib-k}}(t) \theta_{c_{ib-k}}(t)$	(7-3b)	$z_{c_{ib-k}}(t) = \sum_{\substack{ib=1 \\ ib \neq K}}^4 \alpha_{ib-k, ia-K} f_{c_{ib-k}}(t) \theta_{c_{ib-k}}(t)$	(12-3b)
$f_{c+1}(t)$		$f_c(t) \leq n_{c_{ib-k}}(t)$ $f_c(t) \leq Q_c(t)$ $f_c(t) \leq N_c(t) - n_c(t)$ $c \in C_{ia-K}$	
$\zeta(t) = \min\{n_{c_{ib-k}}(t), Q_c(t), N_c(t) - n_c(t)\}$ $\equiv C_{ia-K}$			
$\sum_{m=1}^M \theta_i^m(t) = 1 \quad \forall i \in I$	(8)	$-M \times \theta(t) \leq z(t) \leq M \times \theta(t)$ $-M \times (1 - \theta(t)) \leq f(t) - z(t) \leq M \times (1 - \theta(t))$	(13a) (13b)
		$\sum_{m=1}^M \theta_i^m(t) = 1 \quad \forall i \in I$	(14)
$\sum_{m=1}^{+1} \theta_i^m(t) \geq 0$	(9)	$\sum_{t=1}^{T_m+1} \theta_i^m(t) \geq 0$	(15)

TABLE IV
CASE STUDY INFORMATION

Parameters	
Number of intersections	6
Distance between consecutive intersections (m)	276
Length of each cell (m)	138
Free flow speed (km/h)	50
Number of lanes in each link	2
Saturation flow rate of each link (veh/h)	3600
Time-step duration (sec)	10
Number of time-steps in the optimization problem	3

two different penetration rates of connected vehicles in a random cell is also demonstrated in Fig. 9. Table VI summarizes some summary statistics related to the performance of the NN method for various penetration rates of connected vehicles.

Similar to the results of the Kalman technique, as the penetration rate of connected vehicles increases, the NN

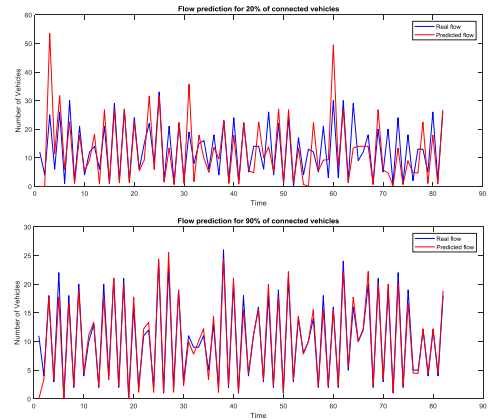


Fig. 8. Estimation of flow for various penetration rates of CVs.

model estimates the residual number of vehicles with more accuracy. Moreover, for the reasonable performance of the Kalman filter and neural network models, at least

TABLE V

COMPARISON OF SUMMARY STATISTICS OF THE KALMAN METHOD FOR VARIOUS PENETRATION RATES OF CVs

Penetration rate	R-squared	RMSE	MAPE
10%	0.5205	8.9841	0.4049
20%	0.6322	7.0880	0.3538
40%	0.6584	3.8721	0.2941
60%	0.8798	2.9354	0.1941
90%	0.9529	1.6723	0.0838

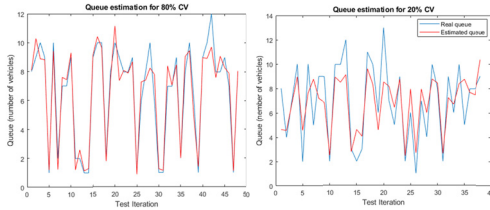


Fig. 9. Comparison of the residual vehicle prediction in a random cell for two different penetration rates of CVs.

TABLE VI

COMPARISON OF THE PERFORMANCE OF THE NN MODEL FOR DIFFERENT PENETRATION RATES OF CVs

Penetration rate	R-squared	RMSE
20%	0.6591	3.4945
40%	0.7883	2.42
60%	0.8613	1.66
80%	0.9533	0.5399

20% of vehicles need to be equipped with communication devices.

Next, the estimation methods along with the control optimization algorithm are used to design the best traffic signal plan which then is implemented in Vissim. We also run the simulation based on the proposed control strategy for a sufficient number of times and then some indexes such as total travel time, delay, and emission are collected for further comparison and analysis. Therefore, the role of Vissim in this paper is twofold: first, it is used as a platform to collect CV data for training the estimation algorithms, and second, the Vissim is utilized for implementing the estimation and optimization methods to collect the traffic data for comparison and analysis. The results are compared for different penetration rates of connected vehicles. Moreover, we test the efficiency of the optimization algorithm using none of the estimation methods (i.e. NN and KF). In this situation, to update the initial condition of cells, we simply consider the number of connected vehicles in each cell to be a rough estimation of the total number of vehicles. An optimized actuated coordinated traffic signal control strategy [60] is considered as a benchmark for this study. This actuated signal plan only uses the loop detector data. since the average vehicle demands are almost the same in corridors, corridors 1 and 2 are considered as major streets with two-way coordination. The cycle length is

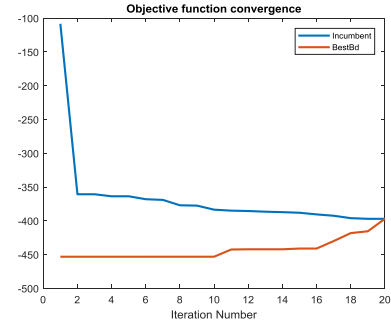


Fig. 10. Incumbent (upper bound value), BestBd (lower bound value).

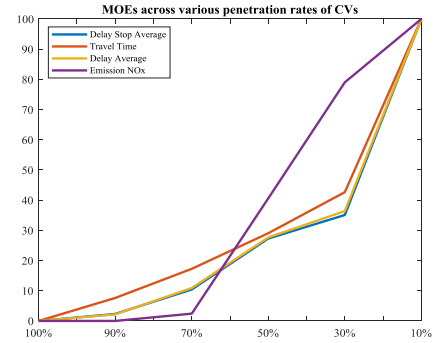


Fig. 11. Comparison of MOEs for various penetration rates of CVs.

76 seconds with the green extension unit of 5 seconds for the minor directions (corridors 3, 4, and 5). The loop detectors are only placed in minor streets. Moreover, It is a very safe assumption that for each phase/movement the maximum green is considered to be half of the cycle time.

For the sake of simplicity, four conflicting movements are considered for each intersection. The optimization algorithm is formulated in MATLAB and solved by Gurobi optimizer on a laptop computer with Core i7-7500U CPU @ 2.7 GHz, 2.9 GHz, and 16 GB RAM. The results of all simulations are obtained in real-time since the duration of each control step is 10 seconds and the computational time of the optimization and estimation algorithms are less than 4 seconds [20], [25]. The stopping criterion used in the optimization algorithm is based on minimizing a relative difference or gap between the upper band and lower band values. The upper bound is the best-found integer feasible solution which is called incumbent and the lower bound is the best relation bound for the best possible objective and called BestBd. The optimization stops whenever the difference between the incumbent and BestBd becomes less than the predetermined gap. The minimum gap between the upper bound value and lower bound values as a termination criterion for the Gurobi optimizer is set to be $1e-4$. (note that the maximization problem is turned to a minimization by multiplying the objective function by minus one). Fig. 10 represents the convergence behavior of the solution for the last 20 iterations such that in the last iteration the gap between the upper bound and lower bound values of the objective is less than $1e-4$.

Fig. 11 shows four performance indexes (delay stop average, travel time, delay average, and emission NOx) recorded for

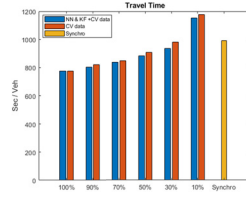


Fig. 12. Total average travel time for different PR of CVs (second).

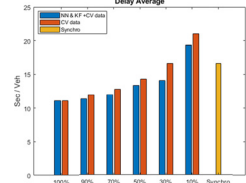


Fig. 13. Average delay for all vehicles for different PR of CVs (second).

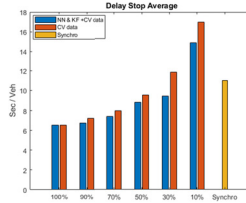


Fig. 14. Average stop delay for all vehicles for different PR of CVs (second).

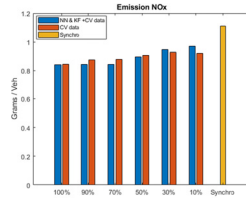


Fig. 15. Total Co and NOx emission for different PR of CVs (grams).

various PRs. For better visualization, the data are normalized to be in the range of 0 to 100. As shown in the figure, as the PR increases the algorithm expectedly performs better. The important point to make is that the improvement for PR greater than 90% is marginal. The results in Table VII, which are the average for ten different simulation runs per scenario also confirm this claim. As can be seen, at the penetration rate of 70% a significant improvement across the above four indexes is gained.

Fig. 12 to Fig. 15 compare the result for different measures of effectiveness (MOEs) across various PRs. For each criterion, we compare the result for with and without estimation techniques. Moreover, the results for different PRs are compared with an optimized actuated-coordinated signal plan. The first criterion is the total average travel time spent by all vehicles while running the Vissim traffic simulation for a specific period. The second metric indicates the average delay per vehicle which is calculated by dividing the total delay to the sum of the number of vehicles. The third one shows the average standstill time per vehicle. This is calculated by

TABLE VII
NUMERICAL RESULTS

	Travel Time		Delay Ave		Delay Stop Ave		Emission NOx	
PR	CV data + NN & KF	CV data	CV data + NN & KF	CV data	CV data + NN & KF	CV data	CV data + NN & KF	CV data
10%	1153.2	1177.4	19.34	21.05	14.83	16.99	0.970	0.919
30%	935.64	979.78	14.11	16.63	9.46	11.89	0.943	0.926
50%	884.07	908.62	13.39	14.33	8.82	9.57	0.893	0.904
70%	839.32	849.88	12.02	12.79	7.43	7.98	0.843	0.875
90%	802.71	821.69	11.3	11.92	6.76	7.23	0.84	0.873
100%	773.87	773.87	11.12	11.12	6.55	6.55	0.84	0.846
actuated	990.77		16.62		11.026		1.11	

dividing the total standstill time delay to the sum of the number of vehicles. The fourth MOE describes the quantity of carbon monoxide (in gram) for the simulation run time. Table VII summarized the numerical results of the adaptive traffic signal control algorithm for with and without prediction algorithms (NN and KF) across different PRs. The results provided in this table are the average for ten different simulation runs with different random seeds for each penetration rate of connected vehicles.

The results are visualized in Fig. 12 to Fig. 15.

As can be seen from the figures, the proposed algorithm in this paper performs better than the actuated-coordinated signal plan for all MOEs provided in table VII for more than 30% penetration rates of connected vehicles. The reasons are twofold: firstly, the estimation algorithms used to estimate the real traffic state from CV data can significantly improve the quality of the input data provided to the adaptive signal controllers and hence result in the accurate performance of the control algorithm. Whereas, the coordinated actuated timing plan purely uses the loop detector data as input. Secondly, the signal control plan in this study is designed based on the rolling horizon strategy which optimizes the signal plan considering some steps ahead of the traffic states. Therefore, it outperforms the actuated coordinated timing plan when a sufficient number of connected vehicles are available (say more than 30% CVs). Almost across all penetration rates of connected vehicles and all MOEs, using the NN and KF estimation methods results in a better performance of the control algorithm in comparison with not using the estimation methods. However, for 10% connected vehicles the results with using the estimation methods are slightly worse than the results for not using these methods. The reason for that could be the poor performance of estimation methods under a very low market penetration rate of connected vehicles. Overall, using the estimation methods in the algorithm yields better performance by 2.2% decrease in total travel time, 7.47% drop in average delay, 10.56% decrease in average stop delay, and less than 1% reduction in NOx emission across the average of the six selected penetration rates of CVs in comparison with not using the estimation methods in the proposed control algorithm.

Furthermore, the proposed algorithm with estimation methods leads to 9.35% reduction in total travel time, 18.49% decrease in average delay, 18.60% decline in average stop

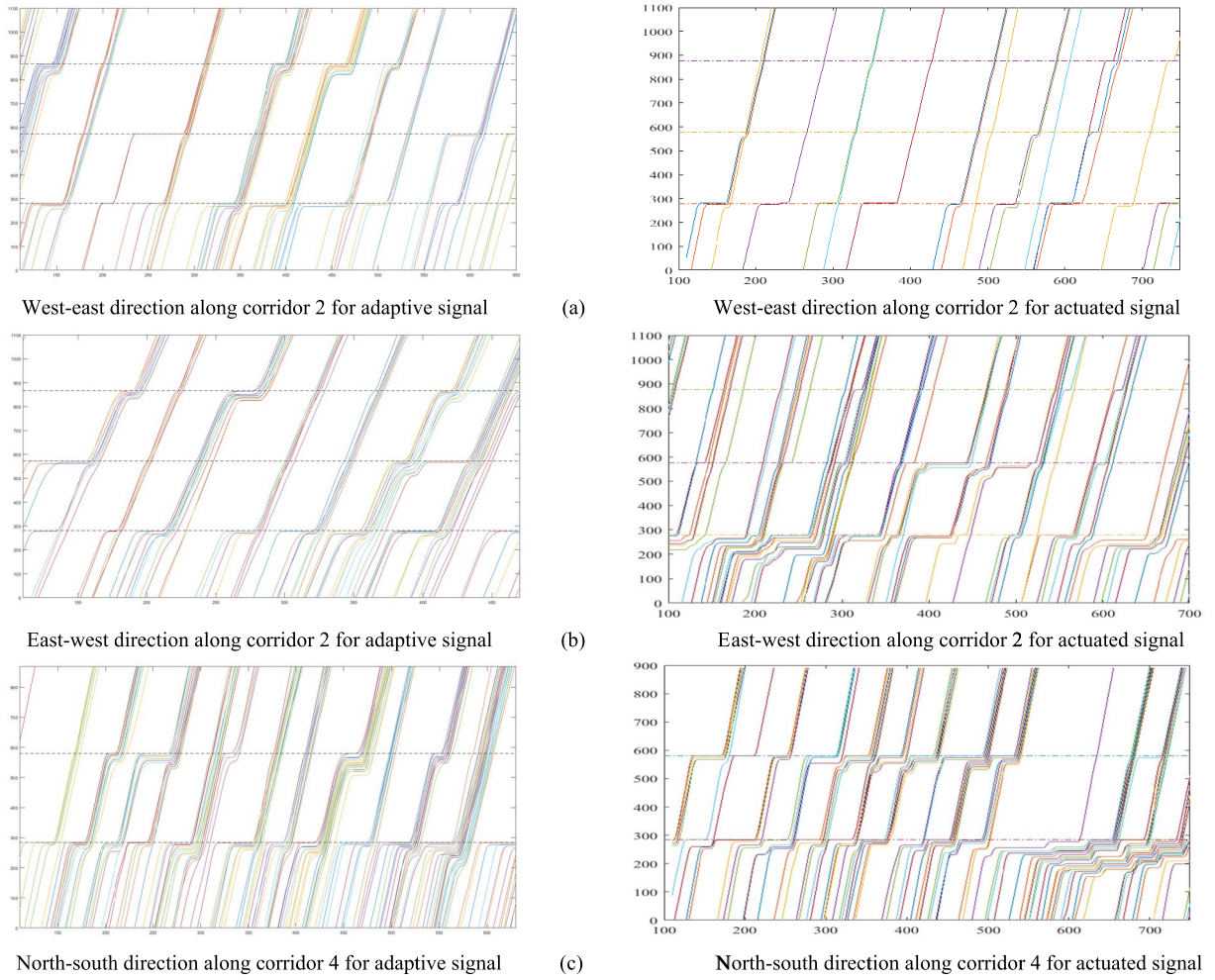


Fig. 16. Comparison of vehicles' trajectory for adaptive and actuated-coordinated signal plan.

delay, and almost 19.83% reduction in NO_x emission on average across all penetration rates of CVs compared to an optimized actuated-coordinated timing plan. These results show the effectiveness of the proposed methodology in comparison with an actuated coordinated signal plan. The proposed method can perform well (in terms of indexes in table VII) even in the situation with a low market penetration rate of CVs. Even the algorithm still has a better performance than the actuated-coordinated signal plan for 30% penetration rate in most of the MOEs.

Fig. 16 illustrates the trajectory of all vehicles in time-space diagrams for some corridors in Fig. 7, for the proposed algorithm with 100% penetration rate of connected vehicles and actuated-coordinated timing setting. This actuated signal plan only uses the loop detector data. The average vehicle demands are almost the same in all corridors (the vehicle demands are subject to change at each time step). However, corridors 1 and 2 are considered as major streets with two-way coordination. The cycle length is 76 seconds with the green extension unit of 5 seconds for the minor directions (corridors 3, 4, and 5). The loop detectors are only placed in minor streets. Based on the actuated-coordinated signal plan in this paper, the coordination is only provided for the intersections

located in corridors 1 and 2. Therefore, vehicles traveling along corridors 1 and 2 in east-west or west-east directions experience less stop time at signalized intersections in comparison with vehicles traveling in other corridors of the network.

As can be seen from the figure, although the rolling horizon control algorithm does not explicitly adjust the offset in the signal plan, it provides coordination in almost all corridors. This coordination results in a few numbers of stops for platoons traveling in the network. The coordination obtained in this algorithm is the result of exchanging information between intersections using connected vehicles. The horizontal dashed lines in Fig. 16 shows the location of intersections. As shown in this figure, most of the vehicles cross the intersections with no stop or a few stops. Although the implementation of the actuated-coordinated signal plan causes the vehicles in corridors 1 and 2 to experience fewer stop time at signalized intersections, (Fig. 16 (a,b)), they face with long stop delay in other corridors (Fig. 16 (c)). Moreover, as can be seen in Fig. 16 (c), some vehicles face the queuing delay as well as the stop delay at signalized intersections. Considering the vehicle trajectory for all corridors, the queuing delay does not occur in our proposed rolling horizon strategy for any corridor. Note that figure 16 only shows the trajectories for some



Fig. 17. Average speed distribution with actuated timing setting.

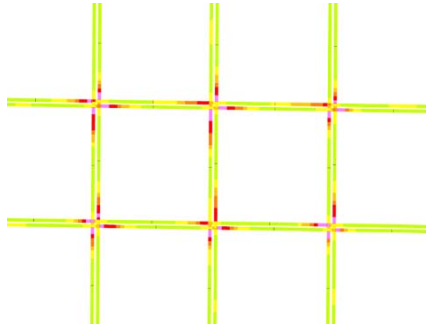


Fig. 18. Average speed distribution with the proposed timing setting (PR = 50%).

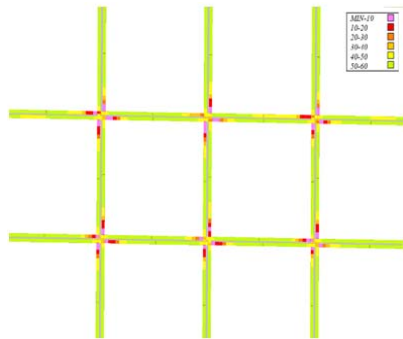


Fig. 19. Average speed distribution with the proposed timing setting (PR = 100%).

sample corridors in a specific time interval. Since the demand is subject to change at each time interval, the demands are not the same for corridors represented in figure 16. However, the average demand for the whole simulation period is almost the same for all corridors.

Fig. 17 to 19 show the average speed of vehicles in each link for actuated signal, our methodology with 50% CV, and our methodology with 100% CV. As it can be seen in the figures, our proposed control algorithm results in higher average speed in comparison to the actuated coordinated signal plan for various PRs of CV. However, we only illustrate the result for 50% and 100%. In this study, the free flow speed is considered to be 50 km/h. The vehicles with the proposed signal setting in this paper travel with the free-flow speed in most parts of the network. However, because of the stop delay and queuing delay experienced by vehicles with

actuated-coordinated timing plan, vehicles have a lower speed than free-flow speed in many parts of the network.

Since the proposed traffic signal algorithm considers the traffic situation in the whole network and does not only rely on the traffic status near the approaches of the intersections (such as actuated signal plans), it can result in better progression of the vehicles in the network. Moreover, since the optimization algorithm considers some steps ahead of the traffic, it prevents the myopic determination of the signal timing and results in the near-optimal solution for the whole control horizon.

VIII. CONCLUSION

In this paper, we propose a method to optimize traffic signal plans based on information gathered from connected vehicles. A novel estimation algorithm is used that incorporates the neural network and Kalman filter methods to estimate the number of vehicles in each cell based on CV data. The cell transmission model is applied as a model in a rolling horizon optimization problem to describe the dynamic of the traffic. We set up a linearized optimization model to maximize intersections throughput. The current state of all cells which is the input for the optimization problem is updated based on the data gathered from estimation methods. The methodology is coded in MATLAB and it calls Gurobi to solve the optimization problem. The results of running the algorithm for a network of six intersections show the superiority of the proposed method over an optimized actuated coordinated traffic signal plan by a significant margin.

Moreover, a comparison between the results provided in this paper demonstrates that the proposed methodology can outperform the actuated signal plan for a range of penetration rates of connected vehicles. Even the results for 30% penetration rate of connected vehicles are still comparable with the results of the actuated coordinated traffic signal plan. In other words, when it comes to the issue of the penetration rate, the proposed algorithm is a robust model that can work accurately (in terms of indexes provided in table VII) even with low percentages of CVs. Moreover, the algorithm can reach its maximum performance even with an imperfect market penetration rate (about 90%). However, As noted before, the main mandate of this research is to set up an effective optimization algorithm for traffic signal settings using the CV data in which a bare minimum PR of 30% is proven to be effective. Nevertheless, a much better flow prediction is still a subject of intensive research, hence expecting a much lower PR is not a far-fetched assumption. It is shown that fusing data from multiple sources of traffic data could be an efficient strategy to overcome the problem of low PRs in traffic estimation algorithms [61]. This concern is thoroughly addressed in one of our papers already published in the “Simulation Modelling Practice and Theory” journal [62]. In this paper, we discussed to fuse the CV data with the data from other existing transport sensors such as Bluetooth sensors to be able to estimate the traffic states such as flow and queue with more accuracy with a limited number of connected vehicles. In this paper, we showed that the traffic state estimation results could be enhanced by integrating the Bluetooth data with CV data in which we confirmed that only 10% of CV is enough to accurately estimate the traffic states.

Therefore, we demonstrated that integrating the Bluetooth data with CV data results in the accurate performance of the estimation algorithms. The Bluetooth sensors are affordable sensors that are already available in many transport networks such as the Melbourne network. The Bluetooth sensors provide information such as vehicle count and travel time using the drivers' smartphones.

Furthermore, as a future direction for this research, the authors are trying to develop the estimation methods based on the trajectory data of connected vehicles which can accurately estimate the traffic states in the situation with low market penetration rates of connected vehicles.

To reduce the number of decision variables, we set up to only decide on whether to give a green time to a particular movement or not. We also use a simple and linear objective function and constraints. Nevertheless, such a simple and intuitive setup can reproduce a variety of signal timing variables including the adjustment of the offsets. The proposed adaptive traffic signal algorithm shows better coordination in comparison with the actuated coordinated timing plan. This explicit offset adjustment is the result of considering the traffic situation in adjacent intersections in the timing plan of each intersection. Furthermore, in comparison with the actuated coordinated timing plan, the average speed across the network is significantly higher with our proposed signal control strategy. Besides in the situation with low market penetration rates (say less than 10%) the actuated coordinated timing plan seems to work better than our proposed traffic signal strategy. It is important to note that the conventional actuated signals are based on expensive loop detector data, whereas, in our proposed method, the only available data are from a limited number of connected vehicles probing in the network. Hence, one can conclude that to shift from loop detector-based control (actuated signal) to CV based control (this study), the market of CV must have reached a minimum level which itself is a subject for further research.

The control strategy in this paper determines the optimal signal plan more frequently based on real data in comparison with conventional rolling horizon approaches. Moreover, the control variables are determined in forms of binary variables. This results in unaffordable computational time for real-time implementation of the algorithm for large networks. Therefore, to make the model applicable for real-time implementation, as a future direction for this research, the network will be decomposed to some subnetworks and the control decisions between subnetworks will be coordinated.

In developing a practical method tailored to large-sized networks for real-time applications, CPU time might be a concern. To address this concern authors are investigating a decomposition method to dissect the size of the problem into several sub-problems to be solved separately and in parallel. Moreover, for the sake of brevity, the pedestrian movements are not considered in this study. However, the proposed method can be easily expanded to accommodate the pedestrian crossing. To do so, we can add two linear constraints to first avoid pedestrian for waiting too long and second to secure enough time for them to cross the intersection. This is left for future research.

As future directions for this research, we would examine how to consider the requirements of various modes of transport such as emergency vehicles, heavy vehicles [63] and public transports in signal timings. Moreover, the performance of the proposed methodology can be scrutinized for oversaturated traffic situations. Furthermore, one could think of improving the accuracy of estimation methods under low penetration rates. For instance, an additional source of information such as Bluetooth devices or video cameras can be used as an additional input to compensate for the poor performance of estimation algorithms in the traffic situation with a low market share of connected vehicles. In [62], the authors proposed fusing the CV data with the data from other existing sources/sensors such as Bluetooth to be able to estimate the traffic states such as flow with more accuracy when the PR is low. We showed that the traffic state estimation results could be enhanced by integrating the Bluetooth data with CV data in which we confirmed that only 10% of CV is enough to accurately estimate the traffic states.

The turn and park vehicles are not considered in this research since the proposed methodology in this paper has an aggregated approach and does not consider the individual maneuver of vehicles. As the future direction for this research, we will also consider the turn and park.

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